EXPERIMENT - 5

AIM: To create SQL views for department salary summary and employee hierarchy, test the updatability of views, and implement a recursive CTE to display reporting chains.

SOURCE CODE:

```
CREATE TABLE IF NOT EXISTS Department
  ( Dept_ID INT PRIMARY KEY,
  Dept_Name VARCHAR(100) NOT NULL
);
CREATE TABLE IF NOT EXISTS Employee
  ( Emp_ID INT PRIMARY KEY,
  Emp_Name VARCHAR(100) NOT NULL,
  Job_Title VARCHAR(100),
  Salary DECIMAL(10, 2),
  Dept_ID INT,
  FOREIGN KEY (Dept_ID) REFERENCES Department(Dept_ID)
);
INSERT IGNORE INTO Department (Dept_ID, Dept_Name) VALUES (1,
  'Engineering'),
  (2, 'Sales'),
  (3, 'Human Resources'),
  (4, 'Finance'),
  (5, 'Operations');
INSERT IGNORE INTO Employee (Emp_ID, Emp_Name, Job_Title, Salary, Dept_ID) VALUES
  (1, 'Ava Patel', 'Developer', 60000, 1),
  (2, 'Liam Chen', 'Developer', 62000, 1),
  (3, 'Mia Garcia', 'Analyst', 58000, 1),
  (4, 'Noah Smith', 'Engineering Manager', 85000, 1),
  (5, 'Olivia Jones', 'Developer', 61000, 1),
  (6, 'Ethan Brown', 'Analyst', 56000, 1),
```

(7, 'Sophia Wilson', 'Sales Representative', 52000, 2),

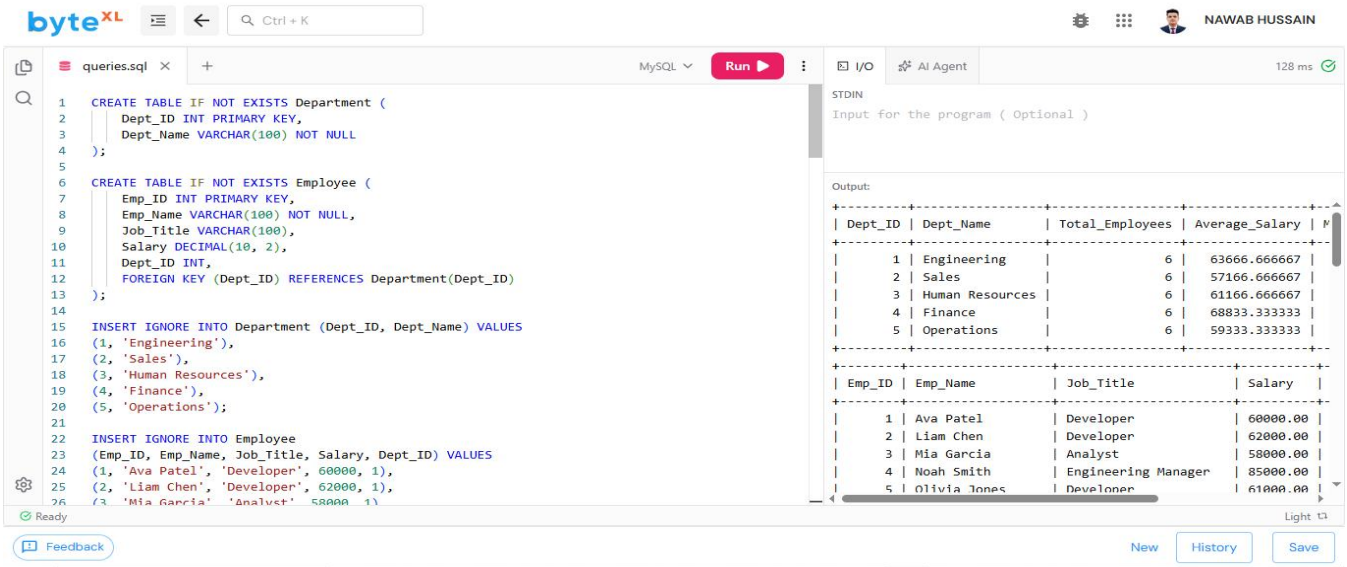
```
    (8, 'Lucas Davis', 'Sales Representative', 54000, 2),
    (6, 'Amelia Miller', 'Sales Manager', 78000, 2),
    (10, 'James Taylor', 'Sales Representative', 51000, 2),
    (11, 'Isabella Anderson', 'Sales Representative', 53000, 2),
    (12, 'Benjamin Thomas', 'Sales Representative', 55000, 2),
    (13, 'Charlotte Moore', 'HR Specialist', 57000, 3),
    (14, 'Elijah Martin', 'HR Specialist', 56000, 3),
    (15, 'Harper Lee', 'Recruiter', 56000, 3),
    (16, 'Daniel White', 'HR Manager', 80000, 3),
    (17, 'Evelyn Harris', 'HR Specialist', 58000, 3),
    (18, 'Henry Clark', 'Recruiter', 60000, 3),
    (16, 'Abigail Lewis', 'Accountant', 63000, 4),
    (20, 'Sebastian Walker', 'Accountant', 64000, 4),
    (21, 'Emily Hall', 'Finance Manager', 88000, 4),
    (22, 'Jack Allen', 'Accountant', 65000, 4),
    (23, 'Ella Young', 'Financial Analyst', 67000, 4),
    (24, 'Owen King', 'Financial Analyst', 66000, 4),
    (25, 'Grace Wright', 'Operations Specialist', 55000, 5),
    (26, 'Matthew Scott', 'Operations Specialist', 56000, 5),
    (27, 'Chloe Green', 'Operations Manager', 82000, 5),
    (28, 'Wyatt Baker', 'Operations Specialist', 57000, 5),
    (26, 'Lily Adams', 'Coordinator', 52000, 5),
    (30, 'Gabriel Nelson', 'Coordinator', 54000, 5);
CREATE VIEW Department_Salary_Summary AS
SELECT
    D.Dept_ID,
    D.Dept_Name,
    COUNT(E.Emp_ID) AS Total_Employees,
    AVG(E.Salary) AS Average_Salary,
```

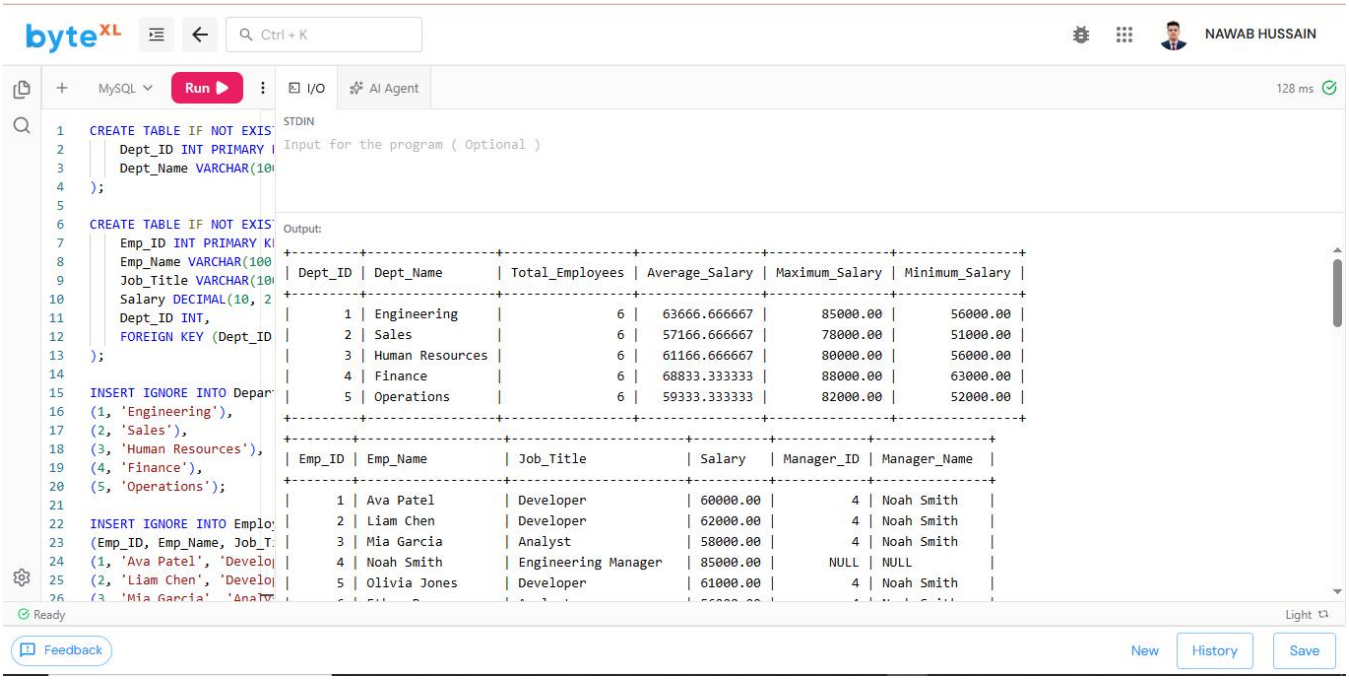
```
  MAX(E.Salary) AS Maximum_Salary,
  MIN(E.Salary) AS Minimum_Salary
FROM Department D
LEFT JOIN Employee E
  ON D.Dept_ID = E.Dept_ID
GROUP BY D.Dept_ID, D.Dept_Name;
SELECT *
FROM Department_Salary_Summary;
ALTER TABLE Employee
ADD Manager_ID INT NULL;
ALTER TABLE Employee
ADD CONSTRAINT FK_Employee_Manager
FOREIGN KEY (Manager_ID)
REFERENCES Employee(Emp_ID);
UPDATE Employee
SET Manager_ID = 4
WHERE Emp_ID IN (1, 2, 3, 5, 6);
UPDATE Employee
SET Manager_ID = 6
WHERE Emp_ID IN (7, 8, 10, 11, 12);
UPDATE Employee
SET Manager_ID = 16
WHERE Emp_ID IN (13, 14, 15, 17, 18);
UPDATE Employee
SET Manager_ID = 21
WHERE Emp_ID IN (16, 20, 22, 23, 24);
UPDATE Employee
SET Manager_ID = 27
WHERE Emp_ID IN (25, 26, 28, 26, 30);
```

```
CREATE VIEW Employee_Hierarchy AS
SELECT
  E.Emp_ID,
  E.Emp_Name,
  E.Job_Title,
  E.Salary,
  E.Manager_ID,
  M.Emp_Name AS Manager_Name
FROM Employee E
LEFT JOIN Employee M
  ON E.Manager_ID = M.Emp_ID;
SELECT *
FROM Employee_Hierarchy;
UPDATE Employee
SET Salary = 60000
WHERE Emp_ID = 1;
SELECT
  Emp_ID,
  Emp_Name,
  Salary
FROM Employee
WHERE Emp_ID = 1;
WITH RECURSIVE Employee_Chain AS
(
  SELECT
    Emp_ID,
    Emp_Name,
    Manager_ID,
    1 AS Level
```

```
  FROM Employee
  WHERE Manager_ID IS NULL
  UNION ALL
  SELECT
    E.Emp_ID,
    E.Emp_Name,
    E.Manager_ID,
    EC.Level + 1
  FROM Employee E
  INNER JOIN Employee_Chain EC
    ON E.Manager_ID = EC.Emp_ID
)
SELECT
  Emp_ID,
  Emp_Name,
  Manager_ID,
  Level
FROM Employee_Chain
ORDER BY Level, Emp_ID;
```

OUTPUT -





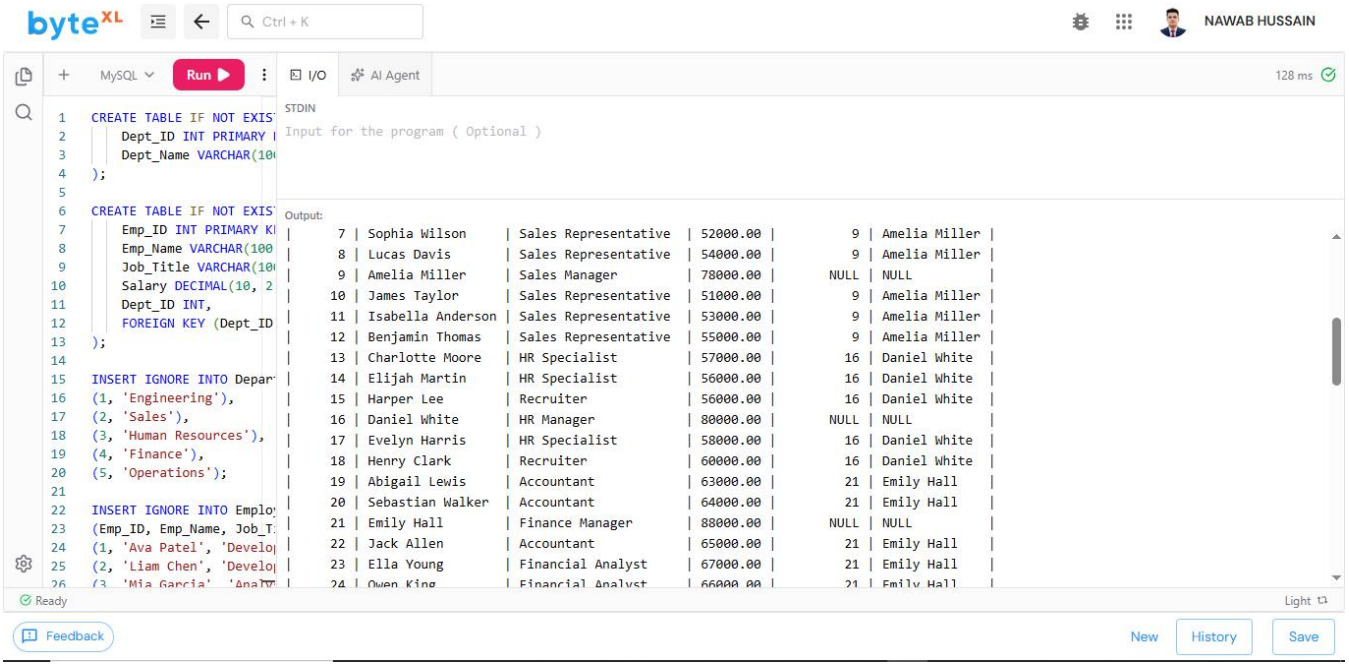

```
CREATE TABLE IF NOT EXIS
    Dept_ID INT PRIMARY
    Dept_Name VARCHAR(10
);

CREATE TABLE IF NOT EXIS
    Emp_ID INT PRIMARY KI
    Emp_Name VARCHAR(100
    Job_Title VARCHAR(10
    Salary DECIMAL(10, 2
    Dept_ID INT,
    FOREIGN KEY (Dept_ID
);

INSERT IGNORE INTO Depar
(1, 'Engineering'),
(2, 'Sales'),
(3, 'Human Resources'),
(4, 'Finance'),
(5, 'Operations');

INSERT IGNORE INTO Emplo
(Emp_ID, Emp_Name, Job_T
(1, 'Ava Patel', 'Develo
(2, 'Liam Chen', 'Develo
(3, 'Mia Garcia', 'Analy
```

```
+---------+-----------------+-----------------+---------------+---------------+---------------+
| Dept_ID | Dept_Name       | Total_Employees | Average_Salary | Maximum_Salary | Minimum_Salary |
+---------+-----------------+-----------------+---------------+---------------+---------------+
|       1 | Engineering     |               6 |   63666.666667 |       85000.00 |       56000.00 |
|       2 | Sales           |               6 |   57166.666667 |       78000.00 |       51000.00 |
|       3 | Human Resources |               6 |   61166.666667 |       80000.00 |       56000.00 |
|       4 | Finance         |               6 |   68833.333333 |       88000.00 |       63000.00 |
|       5 | Operations      |               6 |   59333.333333 |       82000.00 |       52000.00 |
+---------+-----------------+-----------------+---------------+---------------+---------------+
+--------+------------------+----------------------+----------+------------+--------------+
| Emp_ID | Emp_Name         | Job_Title            | Salary   | Manager_ID | Manager_Name |
+--------+------------------+----------------------+----------+------------+--------------+
|      1 | Ava Patel        | Developer            | 60000.00 |          4 | Noah Smith   |
|      2 | Liam Chen        | Developer            | 62000.00 |          4 | Noah Smith   |
|      3 | Mia Garcia       | Analyst              | 58000.00 |          4 | Noah Smith   |
|      4 | Noah Smith       | Engineering Manager  | 85000.00 |       NULL | NULL         |
|      5 | Olivia Jones     | Developer            | 61000.00 |          4 | Noah Smith   |
```




```
CREATE TABLE IF NOT EXIS
    Dept_ID INT PRIMARY
    Dept_Name VARCHAR(10
);

CREATE TABLE IF NOT EXIS
    Emp_ID INT PRIMARY KI
    Emp_Name VARCHAR(100
    Job_Title VARCHAR(10
    Salary DECIMAL(10, 2
    Dept_ID INT,
    FOREIGN KEY (Dept_ID
);

INSERT IGNORE INTO Depar
(1, 'Engineering'),
(2, 'Sales'),
(3, 'Human Resources'),
(4, 'Finance'),
(5, 'Operations');

INSERT IGNORE INTO Emplo
(Emp_ID, Emp_Name, Job_T
(1, 'Ava Patel', 'Develo
(2, 'Liam Chen', 'Develo
(3, 'Mia Garcia', 'Analy
```

```
|      7 | Sophia Wilson     | Sales Representative  | 52000.00 |          9 | Amelia Miller |
|      8 | Lucas Davis       | Sales Representative  | 54000.00 |          9 | Amelia Miller |
|      9 | Amelia Miller     | Sales Manager         | 78000.00 |       NULL | NULL          |
|     10 | James Taylor      | Sales Representative  | 51000.00 |          9 | Amelia Miller |
|     11 | Isabella Anderson | Sales Representative  | 53000.00 |          9 | Amelia Miller |
|     12 | Benjamin Thomas   | Sales Representative  | 55000.00 |          9 | Amelia Miller |
|     13 | Charlotte Moore   | HR Specialist         | 57000.00 |         16 | Daniel White  |
|     14 | Elijah Martin     | HR Specialist         | 56000.00 |         16 | Daniel White  |
|     15 | Harper Lee        | Recruiter             | 56000.00 |         16 | Daniel White  |
|     16 | Daniel White      | HR Manager            | 80000.00 |       NULL | NULL          |
|     17 | Evelyn Harris     | HR Specialist         | 58000.00 |         16 | Daniel White  |
|     18 | Henry Clark       | Recruiter             | 60000.00 |         16 | Daniel White  |
|     19 | Abigail Lewis     | Accountant            | 63000.00 |         21 | Emily Hall    |
|     20 | Sebastian Walker  | Accountant            | 64000.00 |         21 | Emily Hall    |
|     21 | Emily Hall        | Finance Manager       | 88000.00 |       NULL | NULL          |
|     22 | Jack Allen        | Accountant            | 65000.00 |         21 | Emily Hall    |
|     23 | Ella Young        | Financial Analyst     | 67000.00 |         21 | Emily Hall    |
|     24 | Owen King         | Financial Analyst     | 66000.00 |         21 | Emily Hall    |
```